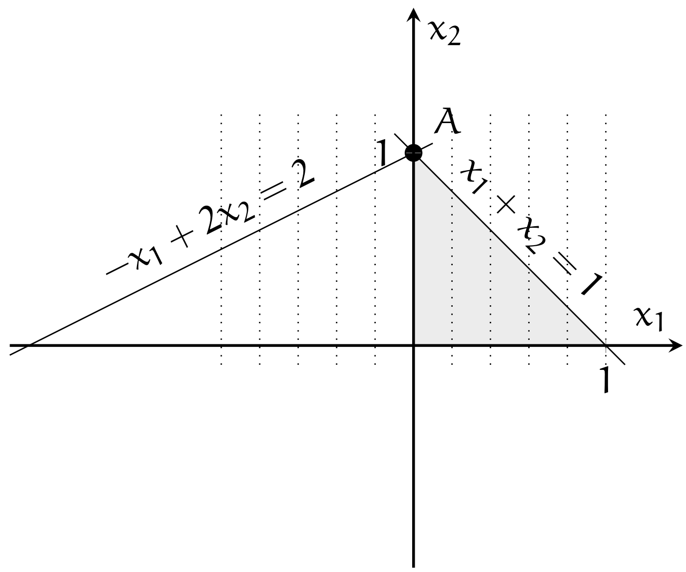
Consider the linear optimization problem

$$\min x_1$$

subject to

$$\begin{aligned} x_1 + x_2 &\leq 1, \\ -x_1 + 2x_2 &\leq 2, \\ x_1, x_2 &\geq 0. \end{aligned}$$



In standard form, after introducing the two slack variables x_3 and x_4, we have

$$\begin{aligned} x_1 + x_2 + x_3 &= 1, \\ -x_1 + 2x_2 + x_4 &= 2, \\ x_1, x_2, x_3, x_4 &\geq 0. \end{aligned}$$

The vertex A corresponds to $x_1 = 0$, $x_2 = 1$, $x_3 = 0$, $x_4 = 0$, and is degenerate. Consider the basis where the variables x_1 and x_2 are in the basis.

1. Calculate all basic directions and represent them on the picture. Calculate the slope of the objective function along each of them.

2. Calculate all the reduced costs.